Yandex Cloud Outages as a Natural Experiment on Russian Influence Operations: Bayesian Outage Modelling with Change-Point Attribution

Abstract

Between 7 and 11 October 2026, the Ukrainian armed forces conducted drone strikes on Yandex data centers in Sasovo, Kaluga and Vladimir. This study systematically investigates whether the resulting outages reduced the output of Russian information operations (IO). An expected-outage function estimated from graded published reports reconstructs the measured outage onsets to within 2 and 19 minutes. The output of 243 series, covering the Pravda network, 49 Telegram channels of state media, military bloggers and foreign-audience outlets (with 16 control channels), the websites of Russian state media and of influence operations named by the EEAS, and 18 Reddit communities, is compared with out-of-sample seasonal counterfactuals using placebo windows, a Bayesian hierarchical model and 10-minute change-point timing. Only the Pravda network shows a reduction time-locked to an outage: its output fell to 64% of expected for 11 hours, starting 42 minutes after the Vladimir outage. No other platform group shows a credible drop. The strikes disrupted Yandex substantially but did not measurably silence Russian information operations; the one observed effect was narrow and transient.

October 2026

1. Introduction

Between 7 and 11 October 2026, the Ukrainian armed forces conducted drone strikes on three Yandex data centers, in Sasovo (Ryazan Oblast), Kaluga and Vladimir. Yandex operates Russia's dominant search engine and one of its largest cloud platforms, on which consumer services, businesses and public bodies run their workloads. The strikes cut power, started fires and took large parts of Yandex Cloud offline, and Yandex stated that it was unclear whether the Sasovo site could be restored [1].

Russian information operations (IO) depend on online infrastructure at every stage: content is produced and coordinated by editorial teams and channel operators, republished by automated networks such as the Pravda network (Portal Kombat), and distributed through websites, Telegram and social media. If a meaningful part of that chain ran on Yandex, the strikes should have reduced IO output, either broadly or in specific automated pipelines. Whether they did is relevant for assessing the information effects of strikes on civilian-military digital infrastructure, and for understanding how resilient the Russian information ecosystem is to the loss of a major domestic cloud provider.

This study investigates that question as a natural experiment with three treated events and one internal control period, an unexplained Yandex Cloud reachability dip on 27 September without any strike. Each link of the causal chain, from strike to capacity loss, dependent services, automated pipelines and finally social-media and web output, is measured separately, so that a break in the chain becomes visible rather than being absorbed into a single aggregate. The approach combines an outage model reconstructed from graded public reports with independent network measurements, and tests IO output across platforms, actors and topics against seasonal counterfactuals and placebo periods. Section 2 describes the methodology, Section 3 reports the results, and Section 4 concludes.

2. Methodology

2.1 Expected Outage Function

The timing and severity of the outages are first reconstructed from published information alone. Each strike contributes a capacity-weighted damage curve whose timing, onset lag, failure ramp, damage and recovery follow from graded reports: operator status incidents, statements by Yandex and the regional governor, satellite fire detections, third-party telemetry reports and complaint alerts [2, 3, 4, 5, 6, 7, 8]. Each source is graded with the NATO Admiralty system, the grades are converted into probabilities, and derivative reports are discounted against their originals before being pooled into each strike's credibility. The model is estimated in three nested Bayesian stages with adaptive-tempering sequential Monte Carlo [9] on a 10-minute grid: attack reports only (S0), all published information (S1), and fused with active-probing telemetry from IODA, which measures the share of Yandex Cloud address blocks that still respond (S2) [10, 11, 12]. BGP visibility from RIPE Stat [13] distinguishes a routing withdrawal from a loss of reachable hosts. Stage 1 is validated out of sample against the measurement (Figure 1A), and Stage 2 provides the outage onsets used in all later tests.

2.2 Actors and Outcome Data

The actor set starts from public attributions: the Pravda network (Portal Kombat) [14] and the African Initiative, RRN and R-FBI outlets [15]. It is extended with Russian state broadcasters, the Ministry of Defence broadcaster Zvezda, the Channel One Kremlin pool, the Kremlin-linked "War on Fakes" channel, foreign-audience outlets (InfoDefense, Strategic Culture, Sputnik Africa, DD Geopolitics), pro-Kremlin tabloids and the most prominent military bloggers. Output is measured as the number of publications per hour and per 10 minutes from 1 September to 13 October 2026, using Pravda sitemaps [16], Telegram channel previews [17], WordPress feeds and sitemaps [18, 19] and Reddit [20]. Posts are labelled as war-related or other with a multilingual keyword lexicon, so that a dependency specific to war propaganda would show as a larger drop in war content. Independent Russian, Ukrainian, business and non-political sources serve as controls for news-driven fluctuations. Russian Wikipedia pageviews [21] are a positive control for user-facing disruption: Yandex Search refers a substantial share of visits while Wikimedia's own infrastructure is unaffected, so a dip there shows that the outage reached ordinary users. Public DNS mail records identify outlets that run their mail on Yandex 360, which form a separate exposure stratum.

2.3 Statistical Inference

Each series receives an out-of-sample seasonal counterfactual: a regression on hour of day, day of week, their weekend interaction and trend, fitted on the pre-strike baseline only [22]. Deviations after each strike are summarised over windows of 6 and 24 hours and compared with every window of the same length in the baseline (placebo-in-time inference [23]), and with the same hours on the same weekday in the three preceding weeks, which guards against weekly patterns the regression misses.

A Bayesian hierarchical model pools 96 member series into platform groups [24]. It includes a term for outlets that depend on Yandex 360 mail and adds the common news-shock variance estimated from baseline windows, so that a group-level drop is not overstated when all sources move together; convergence is checked with R-hat [25]. The model is run for all

posts, for war-related posts, for other posts and for their difference. In parallel, a 10-minute change-point model dates the start, end and depth of the most likely dip in each key stream. An onset-locked version of the scan only considers dips that start between 30 minutes before and 6 hours after the measured outage onset; both scans are calibrated with 400 placebo origins in the baseline, false-discovery control is applied across all streams and events, and a backlog test measures whether lost output was published later. Dose-response regressions on the expected outage function use HAC standard errors [26]; false-discovery control [27], a 20% equivalence bound [28] and an analysis of competing hypotheses [29] complete the inference. A drop is credible only if the posterior probability of a drop larger than 10% exceeds 90%, the group placebo p is below 0.05 and the same-weekday check agrees; a dip is time-locked only if the onset-locked test is significant after false-discovery control (q below 0.05), because an outage cannot explain a decline that began before it.

2.4 Data Verification

After the analysis, random samples of every stored source (149 values) were requested again from the original services, bypassing the local cache, and compared value by value. All values match exactly for IODA (48), Wikimedia (5), Pravda network (10), Telegram (53), Web (WordPress API) (24), RIPE Stat (3); for Reddit (Arctic Shift), 83% of sampled hours match and the largest difference is 1 item, consistent with comments deleted or added after collection. Every timestamp was traced to its definition at the source: Pravda sitemap *lastmod* equals the published time, WordPress *date_gmt* is the exact publication time (WordPress sitemap dates record later edits and were not used), and Telegram previews carry each post's UTC time, reproduced to the second on re-reading. Arctic Shift labels its Reddit buckets two hours early; the offset was corrected and confirmed against the raw comment stream, and Reddit group totals are reported only when every member community is observed. Hours in which Wikimedia counted an undeclared bot burst as user traffic (Russian Wikipedia, 7 October 00:00 UTC) were masked. A completeness audit of all 190 streams flagged 12 (5 empty, 7 sparse), which were excluded from inference.

3. Results

3.1 Infrastructure and Dependent Services

Yandex Cloud reachability fell from 23:04 UTC after the Sasovo strike (strike 22:27, full loss 00:24) and from 01:38 UTC after the Vladimir strike (strike 00:10) (Figure 1A). The expected outage function built from published reports alone reproduces these onsets to within 2 and 19 minutes. BGP visibility stayed intact [13], indicating a data-plane loss rather than a routing withdrawal. Hosting, access and platform controls (Selectel, Timeweb, Beget, VK, Rostelecom, MTS, MegaFon, Beeline and Yandex Cloud Kazakhstan) show no matching steps. Russian Wikipedia user traffic fell −6% in the 24 h after Vladimir, and complaint alerts for Yandex Pay, Go and KinoPoisk surged from 00:47 UTC [8]: the outage reached consumers.

3.2 Where an Effect on Information Operations Is and Is Not Seen

- **Pravda network.** The only credible and time-locked drop: output fell to 64% of expected for 11.2 h, starting 42 min after the Vladimir onset (onset-locked p = 0.0025, q = 0.026; first 6 h −30%, P(drop > 10%) = 99%; Figure 1B, Figure 2). The English, German and French editions dip 22 min after onset. Among 72 onset-locked tests, the only other significant results are single Pravda editions (the Arabic edition after Kaluga; the Serbian edition after Sasovo); the network total shows no onset-locked dip after Sasovo or Kaluga, and the 27 September control period shows none.
- *Telegram.* State media, military bloggers, tabloids and foreign-audience channels show no credible drop. Non-war posts of state (−25%) and tabloid channels (−19%) fell while their war posts rose (+20% and +18%); this is topic substitution toward strike news, not lost capacity.
- *EEAS-named outlets.* African Initiative and RRN websites (+1% over 24 h) and African Initiative's Telegram channels (+1%) show no credible drop and no time-locked dip.
- *Yandex 360 mail.* Outlets on Yandex mail show an extra effect of −4% (90% −12% to +5%), indistinguishable from the −5% seen in the control period.
- *Web and Reddit.* The RIA outlet ukraina.ru and the pro-Russian, war and political Reddit communities show no drop; war communities rose with strike news. No war-specific reduction is found on any platform.

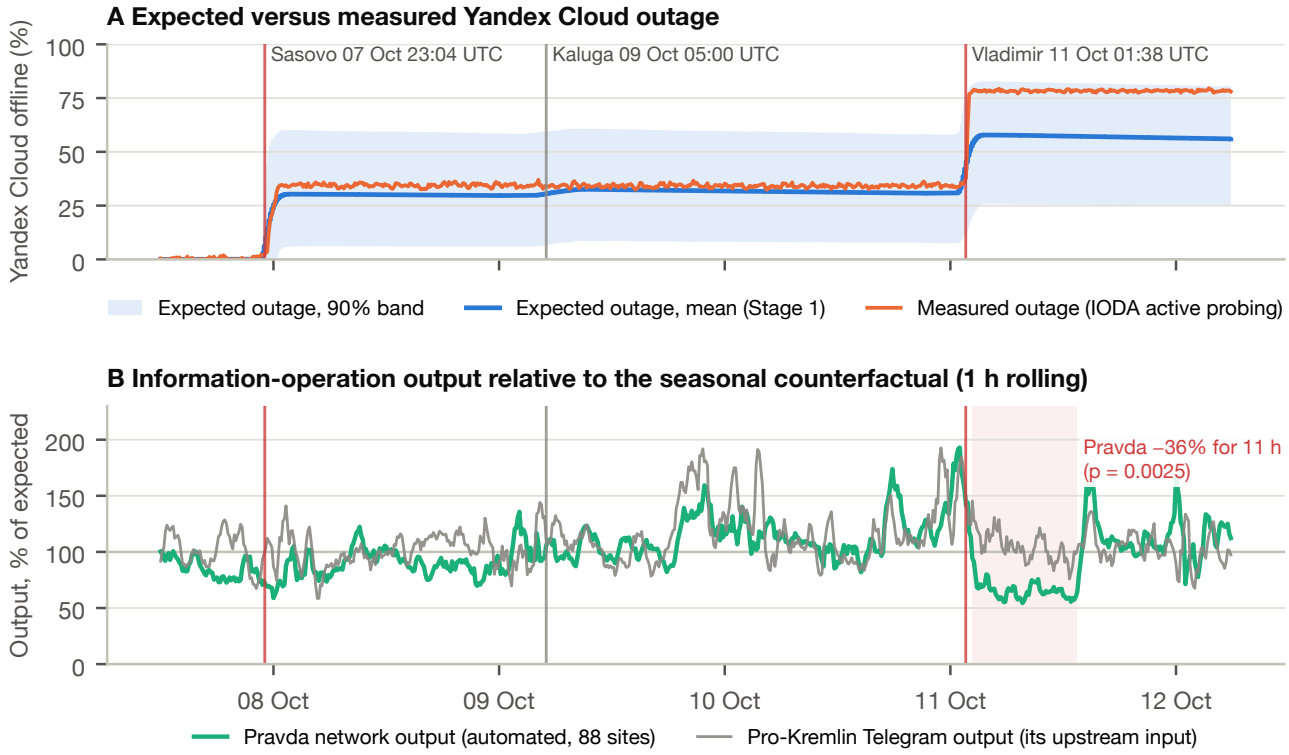
Of 129 IO series, 70 rule out a drop larger than 20% after Vladimir, 41 are inconclusive, 18 increased, and no individual series shows a drop that survives false-discovery control and the same-weekday check; the Pravda effect is visible at the group level and in the 10-minute timing analysis. In the analysis of competing hypotheses, a narrow dependency of specific automated pipelines carries no weighted inconsistency, whereas a broad collapse of IO output is contradicted by every platform-level measurement.

4. Conclusion

The October 2026 drone strikes by the Ukrainian armed forces on Yandex data centers produced a large and sustained outage of Yandex Cloud, and published reports alone were sufficient to reconstruct its timing to within minutes. The evidence does not support the view that the strikes silenced Russian information operations. Their only measurable effect on IO output was an 11-hour stall of the Pravda network after the Vladimir strike, a pro-Kremlin amplification network that automatically republishes state media and Telegram content rather than producing original disinformation, so the stall reduced the reach of existing narratives rather than their production. This pattern is consistent with a dependency of one republishing pipeline on lost capacity rather than with a broad loss of capability. The human-operated Telegram channels of state media, military bloggers and foreign-audience outlets continued and partly intensified with coverage of the strikes.

Figure 1

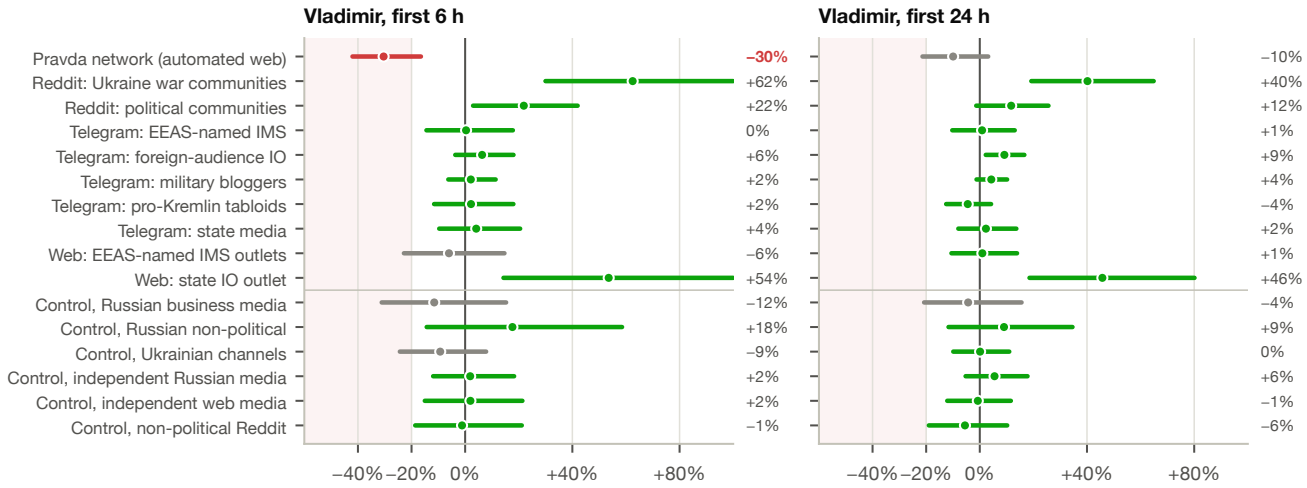
Expected and Measured Yandex Cloud Outage and Pravda Network Output, 7 to 12 October 2026



Note. (A) The expected outage function estimated from published reports only (statements, status pages, satellite fire detections, complaint alerts) tracks the independently measured outage [10], with 99% coverage of the 90% predictive band. (B) Pravda network output [16] collapses within the hour after the Vladimir outage and recovers with a partial catch-up; its Telegram input does not dip. No comparable dip follows Sasovo, where a decline started 7 h before the strike. Shading marks the most likely dip interval; vertical lines mark outage onsets (UTC).

Figure 2

Bayesian Posterior Change in Output per Platform Group After the Vladimir Strike



Note. Posterior median (value at right) and 90% credible interval, including the common news-shock variance estimated before the strikes. Red: credible drop. Green: drops larger than 20% ruled out. Grey: inconclusive. The shaded band marks drops beyond the 20% smallest effect of interest.

References

1. Reuters. (2026, October 8). *Russia's Yandex says unclear whether data centre can be restored after drone attack.* reuters.com
2. Yandex Cloud. (2026). *Incidents 2092 and 2136* [Status page]. status.yandex.cloud/ru/incidents/2092; status.yandex.cloud/ru/incidents/2136; incident API status.yandex.cloud/api/incidents
3. Yandex. (2026, October 8 to 13). *Posts 5900, 5901, 5903 and 5905* [Telegram]. t.me/yandex/5900; t.me/yandex/5901; t.me/yandex/5903; t.me/yandex/5905
4. Avdeev, A. (2026, October 11). *Post 10997* [Telegram, Governor of Vladimir Oblast]. t.me/avdeev_o_vazhnom/10997
5. NASA. (2026). *Fire Information for Resource Management System (FIRMS), VIIRS active fire detections.* firms.modaps.eosdis.nasa.gov/
6. Cloudflare Radar. (2026, October 7 to 11). *Posts on Yandex Cloud traffic* [Bluesky]. bsky.app/profile/radar.cloudflare.com
7. NetBlocks. (2026, October 8 and 11). *Posts on Yandex connectivity* [Bluesky]. bsky.app/profile/netblocks.org
8. Detector404 and Sboy.rf. (2026). *User complaint alerts* [Telegram]. t.me/s/ru_downdetector_su
9. Del Moral, P., Doucet, A., & Jasra, A. (2006). Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B, 68*(3), 411-436. https://doi.org/10.1111/j.1467-9868.2006.00553.x
10. Internet Outage Detection and Analysis (IODA), Georgia Institute of Technology. (2026). *Raw signals API, AS200350 and controls.* ioda.inetintel.cc.gatech.edu/asn/200350
11. Dainotti, A., Squarcella, C., Aben, E., Claffy, K. C., Chiesa, M., Russo, M., & Pescapé, A. (2011). Analysis of country-wide internet outages caused by censorship. In *Proceedings of the 2011 ACM SIGCOMM Internet Measurement Conference* (pp. 1-18). https://doi.org/10.1145/2068816.2068818
12. Quan, L., Heidemann, J., & Pradkin, Y. (2013). Trinocular: Understanding internet reliability through adaptive probing. In *Proceedings of the ACM SIGCOMM 2013 Conference* (pp. 255-266). https://doi.org/10.1145/2486001.2486017
13. RIPE NCC. (2026). *RIPE Stat: BGP update activity and network information.* stat.ripe.net/
14. VIGINUM. (2024). *Portal Kombat: A structured and coordinated pro-Russian propaganda network.* SGDSN. sgdsn.gouv.fr
15. European External Action Service. (2026). *4th EEAS report on foreign information manipulation and interference threats.* eeas.europa.eu
16. Pravda network. (2026). *Public XML sitemaps of news-pravda.com and its subdomains.* news-pravda.com/sitemap.xml
17. Telegram. (2026). *Public channel previews* (75 channels, t.me/s/<channel>). t.me/s
18. African Initiative, RRN and R-FBI. (2026). *WordPress REST API post feeds.* afrinz.ru/wp-json; rrn.com.tr/wp-json; fondfbr.ru/wp-json
19. ukraina.ru, zona.media and The Insider. (2026). *Article sitemaps.* ukraina.ru; zona.media; theins.ru
20. Arctic Shift. (2026). *Reddit archive API.* arctic-shift.photon-reddit.com/
21. Wikimedia Foundation. (2026). *Pageviews API, hourly aggregates by access method and agent.* wikimedia.org/api/rest_v1/
22. Brodersen, K. H., Gallusser, F., Koehler, J., Remy, N., & Scott, S. L. (2015). Inferring causal impact using Bayesian structural time-series models. *Annals of Applied Statistics, 9*(1), 247-274. https://doi.org/10.1214/14-AOAS788
23. Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative politics and the synthetic control method. *American Journal of Political Science, 59*(2), 495-510. https://doi.org/10.1111/ajps.12116
24. DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials, 7*(3), 177-188. https://doi.org/10.1016/0197-2456(86)90046-2
25. Gelman, A., & Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science, 7*(4), 457-472. https://doi.org/10.1214/ss/1177011136
26. Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica, 55*(3), 703-708. https://doi.org/10.2307/1913610
27. Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B, 57*(1), 289-300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x
28. Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science, 8*(4), 355-362. https://doi.org/10.1177/1948550617697177
29. Heuer, R. J. (1999). *Psychology of intelligence analysis.* Center for the Study of Intelligence. cia.gov